

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the field of computer science dedicated to creating systems capable of performing tasks that normally require human intelligence. These tasks include visual perception, speech recognition, decision-making, natural language understanding, and translation.

AI can be broadly divided into two categories: Narrow AI and General AI. Narrow AI, also called Weak AI, is designed to handle a specific task—such as facial recognition, spam filtering, or playing chess. General AI, also known as Strong AI or Artificial General Intelligence (AGI), refers to a hypothetical machine that can understand and reason across any cognitive task at human level. To date, only Narrow AI has been achieved.

The history of AI dates back to the 1950s. The term was coined by John McCarthy at the Dartmouth Conference in 1956. Early work focused on symbolic reasoning and expert systems. The field experienced periods of optimism followed by 'AI winters' when funding and interest waned.

Machine Learning Fundamentals

Machine Learning (ML) is a subset of AI that enables systems to learn patterns from data without being explicitly programmed. ML algorithms build a mathematical model based on sample data, known as training data, to make predictions or decisions without explicit rules.

There are three main types of machine learning: supervised learning, unsupervised learning, and reinforcement learning. In supervised learning, the algorithm learns from labelled training data, mapping inputs to outputs. In unsupervised learning, the algorithm finds hidden patterns in unlabelled data. In reinforcement learning, an agent learns by interacting with an environment and receiving rewards or penalties.

Common ML algorithms include linear regression, logistic regression, decision trees, random forests, support vector machines, k-nearest neighbours, and neural networks. Each has its own strengths and is suited to different types of problems.

A key concept in ML is the bias-variance tradeoff. A model with high bias underfits the data and makes overly simplistic assumptions. A model with high variance overfits the training data and fails to generalise to new data. Finding the right balance is central to ML model development.

Deep Learning and Neural Networks

Deep Learning is a specialised subfield of ML that uses artificial neural networks with many layers—hence 'deep'. A neural network is composed of neurons (nodes) organised into layers: an input layer, one or more hidden layers, and an output layer. Each connection between neurons has a weight that is adjusted during training.

The most common architectures include Feedforward Neural Networks, Convolutional Neural Networks (CNNs) for image processing, Recurrent Neural Networks (RNNs) for sequential data, and Transformers for natural language processing tasks.

Transformers, introduced in the 2017 paper 'Attention Is All You Need', revolutionised NLP by using self-attention mechanisms. This architecture underpins modern large language models such as GPT, BERT, and LLaMA.

Training deep learning models requires significant computational resources, typically GPUs or TPUs. Frameworks like TensorFlow and PyTorch provide the infrastructure for building and training these models efficiently.

Natural Language Processing

Natural Language Processing (NLP) is the branch of AI focused on the interaction between computers and human language. It enables machines to understand, interpret, and generate human language in a meaningful way.

Key NLP tasks include tokenisation, part-of-speech tagging, named entity recognition, sentiment analysis, machine translation, text summarisation, and question answering. Modern NLP relies heavily on transformer-based language models that are pre-trained on massive text corpora and then fine-tuned for specific tasks.

Large Language Models (LLMs) such as GPT-4, Claude, and LLaMA have demonstrated remarkable capabilities in text generation, reasoning, and code generation. These models are accessed via APIs from providers such as OpenAI, Anthropic, Google, and Groq.

Retrieval-Augmented Generation (RAG) is a technique that combines retrieval of relevant documents with LLM generation. The retriever fetches context from a knowledge base, and the LLM generates an answer grounded in that context. RAG helps reduce hallucinations and provides source attribution.

AI Ethics and Responsible AI

As AI systems become more powerful, ethical considerations are increasingly important. Key concerns include bias and fairness, transparency, privacy, accountability, and the potential for misuse.

Bias in AI systems can arise from biased training data, unfair features, or flawed model design. Mitigating bias requires diverse datasets, regular auditing, and fairness metrics.

Transparency means making AI decision-making processes understandable to humans. Techniques include explainable AI (XAI), attention visualisation, and model documentation.

Responsible AI frameworks, such as the EU AI Act and the NIST AI Risk Management Framework, guide the development and deployment of AI in ways that are safe, fair, and beneficial to society.